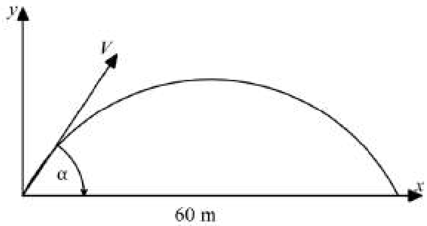
Projectile Motion
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 A cricket player hits a ball from ground level. The ball is projected under gravity with speed $V\ m/s$ in a direction making an angle α with the horizontal. The ball lands on the boundary, which is 60 metres away. 3



The equation of motion of the ball are $x = Vtcos\alpha$ and $y = -\frac{1}{2}gt^2 + Vtsin\alpha$ (do **NOT** prove these), where $g = 9.8\ m/s^2$. If the angle of flight to the horizontal is 28^o at the instant the ball leaves the bat, calculate the initial speed of the ball.

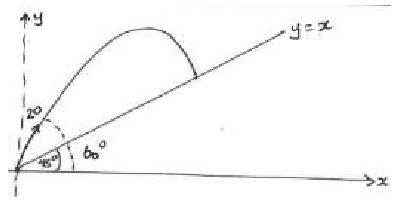
$$x = Vt\cos\alpha$$
$$t = \frac{60}{V\cos 28}$$

$$y = -\frac{1}{2}gt^2 + Vt\sin\alpha$$
$$0 = -\frac{1}{2}g\left(\frac{60}{V\cos 28}\right)^2 + V\left(\frac{60}{V\cos 28}\right)\sin 28$$
$$\frac{g}{2}\left(\frac{60}{V\cos 28}\right)^2 = 60\tan 28$$
$$\left(\frac{60}{V\cos 28}\right)^2 = \frac{120\tan 28}{g}$$
$$V^2\cos^2 28 = \frac{3600g}{120\tan 28}$$
$$V^2 = 709.256...$$
$$V = 26.6\ \text{ms}^{-1}.$$

2 A ball is thrown uphill from the base of a hill inclined at 45^o to the horizontal. The initial velocity was 20 metres per second at 60^o to the horizontal.

The position vector of the centre of the ball at t seconds after release is given by

$\underset{\sim}{s}(t) = 10t\underset{\sim}{i} + (10t\sqrt{3} - 5t^2)\underset{\sim}{j}$ where $\underset{\sim}{i}$ is a unit vector in the horizontal direction of motion of the ball, and $\underset{\sim}{j}$ is a unit vector vertically upward.



(i) Find the Cartesian equation of the ball's path. 1

$$\text{(i)} \quad x = 10t \qquad \therefore t = \frac{x}{10}$$
$$\therefore y = 10\sqrt{3}\left(\frac{x}{10}\right) - 5\left(\frac{x}{10}\right)^2$$
$$\therefore y = \sqrt{3}x - \frac{x^2}{20}$$

(ii) Find its time of flight. 2

ball strikes hill when ball & hill same height,

$\therefore \quad x = \sqrt{3}x - \frac{x^2}{20}$

$\therefore \quad \frac{x^2}{20} + (1-\sqrt{3})x = 0$

$x\left[\frac{x}{20} + 1 - \sqrt{3}\right] = 0$

$\therefore \quad \frac{x}{20} = \sqrt{3} - 1 \quad \therefore \quad x = 20(\sqrt{3}-1)$

Thus $t = \frac{x}{10} \quad \therefore \quad t = 2(\sqrt{3}-1)$ secs

(iii) Calculate the ball's speed when it strikes the hill. 2

(iii) $\dot{x} = 10, \quad \dot{y} = 10\sqrt{3} - 10t$

$= 10\sqrt{3} - 10\left[2(\sqrt{3}-1)\right]$

$= 10\sqrt{3} - 20\sqrt{3} + 20$

$\therefore \dot{y} = 20 - 10\sqrt{3}$

Thus speed $= \sqrt{10^2 + (20-10\sqrt{3})^2}$

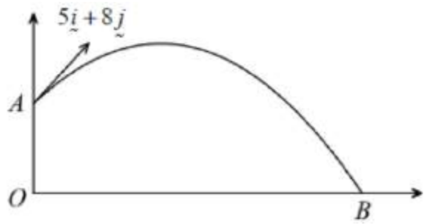
$= \sqrt{10^2 + 10^2(2-\sqrt{3})^2}$

$= 10\sqrt{1+4-4\sqrt{3}+3}$

$= 10\sqrt{8-4\sqrt{3}}$

$= 20\sqrt{2-\sqrt{3}}$ m/s i.e. $\doteqdot 10.35$ m/s.

3 A ball is struck by a racquet from a point A which has position vector $21\underset{\sim}{j}$ metres relative to a fixed origin O. Initially, the velocity vector of the ball is $(5\underset{\sim}{i} + 8\underset{\sim}{j})$ m/s where $\underset{\sim}{i}$ and $\underset{\sim}{j}$ are the unit vectors in the horizontal and vertical directions respectively. After being struck, the ball undergoes projectile motion under the influence of gravity only, that is, the acceleration vector of the ball $\underset{\sim}{a} = -10\underset{\sim}{j}$. The ball will hit the ground at point B, a point that is on the same horizontal level as the origin.



(i) Show that the velocity vector of the ball t seconds after being struck is $\underset{\sim}{v} = 5\underset{\sim}{i} + (8 - 10t)\underset{\sim}{j}$, and hence find an expression for the position vector $\underset{\sim}{r}$ of the ball relative to O at a time t seconds after it was struck. 2

$\underset{\sim}{a} = -g\underset{\sim}{j} = -10\underset{\sim}{j}$

$\underset{\sim}{v} = 5\underset{\sim}{i} - (10t + c_1)\underset{\sim}{j}$ (since horizontal velocity is a constant 5 m/s)

Given $\underset{\sim}{v}(0) = 5\underset{\sim}{i} + 8\underset{\sim}{j}$: $-(10\times 0 + c_1) = 8 \qquad c_1 = -8$

$\underset{\sim}{v} = 5\underset{\sim}{i} - (10t - 8)\underset{\sim}{j} = 5\underset{\sim}{i} + (8 - 10t)\underset{\sim}{j}$

$\underset{\sim}{r} = (5t + c_2)\underset{\sim}{i} + (8t - 5t^2 + c_3)\underset{\sim}{j}$

Given $\underset{\sim}{r}(0) = 21\underset{\sim}{j}$:

$5\times 0 + c_2 = 0 \qquad 8\times 0 - 5\times 0^2 + c_3 = 21$

$c_2 = 0 \qquad c_3 = 21$

$\underset{\sim}{r} = (5t)\underset{\sim}{i} + (8t - 5t^2 + 21)\underset{\sim}{j}$

(ii) Find the maximum height that the ball reaches. 2

Maximum height occurs when $\dot{y}(t) = 0$:

$8 - 10t = 0 \qquad t = 0.8$

$y(0.8) = 8\times 0.8 - 5\times 0.8^2 + 21 = 24.2$ metres

(iii) Will the ball clear a building that is 10 metres away from the origin, and has a height of 16 metres, and if so, by how much? Justify your answer. 1

The position vector of the top of the building is given by $10\tilde{\imath} + 16\tilde{\jmath}$

When $x(t) = 10, \quad 5t = 10 \quad t = 2$

$y(2) = 8(2) - 5(2^2) + 21 = 17 = 16 + 1$

Thus the ball will clear the building by 1 metre.

4 Mick hits a tennis ball from a height 2.8 metres above the ground. The angle of elevation as the racket hits the ball is 5^o and the velocity is 202 km/h. The ball lands at a horizontal distance of d metres from Mick's feet.

(i) Using $g = 9.8\ m/s^2$ as acceleration due to gravity, show that the vector displacement of the tennis ball is: $\underset{\sim}{s}(t) \approx 55.8t\underset{\sim}{i} + (4.9t - 4.9t^2 + 2.8)\underset{\sim}{j}$ 3

initial velocity = 202 km / h

$= 56.\dot{1}$ m / s

$v(0) \approx 56\cos 5° i + 56 \sin 5° j$

$\approx 55.8i + 4.9j$

$\underset{\sim}{\mathbf{a}}(t) = -9.8\underset{\sim}{\mathbf{j}}$

$\underset{\sim}{\mathbf{v}}(t) = \int -9.8\underset{\sim}{\mathbf{j}}\,dt$

$= -9.8t\underset{\sim}{\mathbf{j}} + C$

$\underset{\sim}{\mathbf{v}}(0) = 55.8\underset{\sim}{\mathbf{i}} + 4.9\underset{\sim}{\mathbf{j}}$

$\therefore -9.8(0)\underset{\sim}{\mathbf{j}} + C = 55.8\underset{\sim}{\mathbf{i}} + 4.9\underset{\sim}{\mathbf{j}}$

$C = 55.8\underset{\sim}{\mathbf{i}} + 4.9\underset{\sim}{\mathbf{j}}$

$\underset{\sim}{\mathbf{v}}(t) = -9.8t\underset{\sim}{\mathbf{j}} + 55.8\underset{\sim}{\mathbf{i}} + 4.9\underset{\sim}{\mathbf{j}}$

$\underset{\sim}{\mathbf{s}}(t) = \int 55.8\underset{\sim}{\mathbf{i}} - 9.8t\underset{\sim}{\mathbf{j}} + 4.9\underset{\sim}{\mathbf{j}}\,dt$

$= 55.8t\underset{\sim}{\mathbf{i}} - 4.9t^2\underset{\sim}{\mathbf{j}} + 4.9t\underset{\sim}{\mathbf{j}} + C_1$

$= 55.8t\underset{\sim}{\mathbf{i}} + 4.9t(1-t)\underset{\sim}{\mathbf{j}} + C_1$

$\underset{\sim}{\mathbf{s}}(0) = 2.8\underset{\sim}{\mathbf{j}}$

$\therefore\ C_1 = 2.8\underset{\sim}{\mathbf{j}}$

$\underset{\sim}{\mathbf{s}}(t) = 55.8t\underset{\sim}{\mathbf{i}} + (4.9t - 4.9t^2 + 2.8)\underset{\sim}{\mathbf{j}}$

(ii) If d is greater than 26.2, the ball will be called "out". Find the value of d and determine if the ball will be called "out". 2

When the ball hits the ground its vertical displacement will be zero.

$-4.9t^2 + 4.9t + 2.8 = 0$

$$t = \frac{-4.9 \pm \sqrt{(4.9)^2 - 4(-4.9)(2.8)}}{-2(-4.9)}$$

$t = -0.4$ or 1.41

Ignore negative solution as $t > 0$

$d = s(1.406326967) = 55.8(1.406326967)$

$= 78.47304477$ m

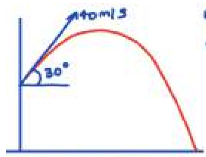
$\therefore d > 26.2$ m

So, the ball will be called out.

5 A particle is projected upwards from the top of a building 12 metres high with an initial velocity of 40 m/s. The particle is launched at an angle of 30 degrees with the horizontal. Take 9.8 m/s^2 for the acceleration due to gravity.

(i) If the origin is at the base of the building, show that the displacement vector of the particle is: 2

$\underset{\sim}{s}(t) = 20\sqrt{3}t\underset{\sim}{i} + (-4.9t^2 + 20t + 12)\underset{\sim}{j}$



when $t = 0$

$v(t) = 40\cos 30\underset{\sim}{i} + 40\sin 30\underset{\sim}{j}$

$= 40 \times \frac{\sqrt{3}}{2}\underset{\sim}{i} + 40 \times \frac{1}{2}\underset{\sim}{j}$

$= 20\sqrt{3}\underset{\sim}{i} + 20\underset{\sim}{j}$

$a(t) = -9.8\underset{\sim}{j}$

$v(t) = \int -9.8\underset{\sim}{j}\,dt$

$= -9.8t\underset{\sim}{j}$

when $t = 0$

$20\sqrt{3}i + 20\underset{\sim}{j} = C$

$\therefore v(t) = -9.8t\underset{\sim}{j} + 20\sqrt{3}\underset{\sim}{i} + 20\underset{\sim}{j}$

$= 20\sqrt{3}\underset{\sim}{i} + (-9.8t + 20)\underset{\sim}{j}$

$s(t) = \int 20\sqrt{3}\underset{\sim}{i} + (-9.8t + 20)\underset{\sim}{j}\,dt$

$= 20\sqrt{3}t\underset{\sim}{i} + (-\frac{9.8}{2}t^2 + 20t)\underset{\sim}{j} + C$

when $t = 0$ $s(t) = 12\underset{\sim}{j}$

$12\underset{\sim}{j} = C$

$\therefore s(t) = 20\sqrt{3}t\underset{\sim}{i} + (-4.9t^2 + 20t)\underset{\sim}{j} + 12\underset{\sim}{j}$

$s(t) = 20\sqrt{3}t\underset{\sim}{i} + (-4.9t^2 + 20t + 12)\underset{\sim}{j}$

(ii) Show that the time of flight is 4.61 seconds, correct to 2 decimal places. 1

The time of flight is when the y component of the displacement is 0.

i.e. $-4.9t^2 + 20t + 12 = 0$

$t = \dfrac{-20 \pm \sqrt{20^2 - 4(4.9)12}}{2(-4.9)}$

$= -0.53$ or 4.61 (But $t > 0$)

$\therefore t = 4.61$ seconds.

(iii) Determine the speed, correct to 2 decimal places, and the angle, correct to the nearest degree, at which the particle hits the ground. 2

iii) $v(t) = 20\sqrt{3}\underset{\sim}{i} + (-9.8t + 20)\underset{\sim}{j}$

when $t = 4.61$

$v(4.61) = 20\sqrt{3}\underset{\sim}{i} + (-9.8 \times 4.61 + 20)\underset{\sim}{j}$

$= 20\sqrt{3}\underset{\sim}{i} + -25.178\underset{\sim}{j}$

$\therefore$ Velocity $= \sqrt{(20\sqrt{3})^2 + (-25.178)^2}$

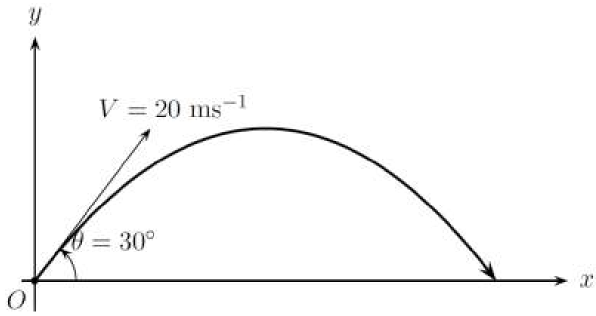
$= 42.83$ m/s

Angle of Inclination: $\tan\theta = \dfrac{-25.178}{20\sqrt{3}}$

$\theta = -36°$

$\therefore$ The particle hits the ground with a velocity of 42.83 m/s at an angle of 144° with the horizontal

6 Eileen throws a rock from a fixed point O on level ground, with velocity $V = 20\ m/s$ at an angle of $\theta = 30^o$ with the horizontal.



The flight path of the rock is given by:

$x_r = Vt\cos\theta$

$y_r = -\frac{1}{2}gt^2 + Vt\sin\theta$ (Do NOT prove this)

where t is the time in seconds after the stone was thrown. (Take $g = 10\ m/s^2$)

(i) Find the maximum height that the rock reaches. 2

Nathan is standing on jungle gym, 3 metres directly above the point O. Nathan throws the stick with velocity $W = 15\ m/s$ at angle α above the horizontal, such that it passes through the point C where the stone reaches its maximum height.

$\dot{y_r} = -gt + V\sin\theta$

$= -10t + 20\sin 30°$

$= -10t + 10$

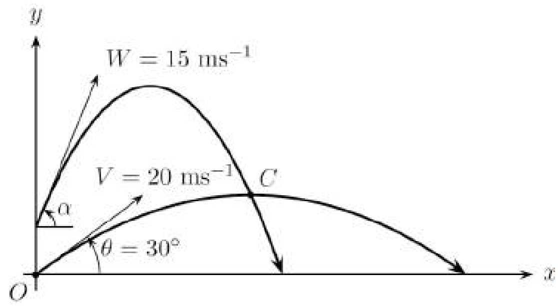
Maximum height occurs when $\dot{y_r} = 0$:

$0 = -10t + 10 \qquad t = 1$

When $t = 1$:

$y_r = -\dfrac{1}{2}(10)(1)^2 + 20(1)\sin 30° = 5$

$\therefore$ The maximum height of the rock is 5 metres.

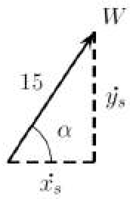


(ii) Prove that the position of stick is given by 3

$r_s = 15tcos\alpha\underset{\sim}{i} + (-5t^2 + 15tsin\alpha{+}3)\ \underset{\sim}{j}$

$$\underset{\sim}{a}_s(t) = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$$

$$\underset{\sim}{v}_s(t) = \begin{pmatrix} c_1 \\ -10t + c_2 \end{pmatrix}$$



When $t = 0$: $\dot{x}_s = 15\cos\alpha$

$\dot{y}_s = 15\sin\alpha$

$$\therefore \underset{\sim}{v}_s(t) = \begin{pmatrix} 15\cos\alpha \\ -10t + 15\sin\alpha \end{pmatrix}$$

$$\underset{\sim}{r}_s(t) = \begin{pmatrix} 15t\cos\alpha + c_3 \\ -5t^2 + 15t\sin\alpha + c_4 \end{pmatrix}$$

When $t = 0$, $x = 0$ and $y = 3$.

$$\therefore \underset{\sim}{r}_s(t) = \begin{pmatrix} 15t\cos\alpha \\ -5t^2 + 15t\sin\alpha + 3 \end{pmatrix}$$

(iii) Determine whether the stick collides with the rock at point C. Justify your answer with calculations. 2

$x_r = 20(1)\cos 30^\circ = 10\sqrt{3}$

If the stick collides with the rock at point C:

when $t = 1$, $x_s = 10\sqrt{3}$

From part (ii), $x = 15t\cos\alpha$:

$$10\sqrt{3} = 15(1)\cos\alpha$$

$$\cos\alpha = \frac{2\sqrt{3}}{3}$$

But $-1 \leq \cos\alpha \leq 1$

$\therefore$ There is no such value of α for which the the horizontal displacement of the stick is $x_s = 10\sqrt{3}$ when $t = 1$.

Hence the rock and stick do not collide at C since they reach the horizontal position of C at different times.

Projectile Motion
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 A projectile is launched at an angle 30^o to the horizontal and at an initial speed of 60 m/s. By taking the point of projection as the origin, the projectile's displacement vector $\underset{\sim}{d}$ at any time t seconds is:

$$\underset{\sim}{d} = (30\sqrt{3}t)\underset{\sim}{i} + (30t - 5t^2)\underset{\sim}{j}$$

(i). Find the maximum height of the projectile. 2

For the maximum height, we need to find the vertical component for velocity (y).
From the given displacement vector, the vertical component (height at any time t) is $30t - 5t^2$.

Differentiate to give: $\dot{y} = 30 - 10t$

Time to reach maximum height occurs when $\dot{y} = 0$.

$\therefore t = 3$ seconds.

$\therefore$ height at $t = 3$ is $30(3) - 5(3)^2 = 45$ metres.

(ii). Find the exact speed of the projectile two seconds after it was launched. 2

When $t = 2$, the horizontal component for velocity $\dot{x} = 30\sqrt{3}$ and the vertical component for velocity $\dot{y} = 10$.

$\therefore$ speed of projectile at $t = 2$ is $= \sqrt{(30\sqrt{3})^2 + (10)^2} = \sqrt{2800} = 20\sqrt{7}$ m/s

2 A particle is projected from the origin with initial velocity 60 m/s at an angle 30° to the horizontal. The acceleration vector is given by 2

$$\underset{\sim}{\ddot{r}}(t) = \begin{pmatrix} 0 \\ -10 \end{pmatrix}, \text{ where } \underset{\sim}{r}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

is the position vector. What are the initial conditions?

$\dot{x} = 30\sqrt{3}, \ddot{x} = 0, \dot{y} = 30$ and $\ddot{y} = -10$

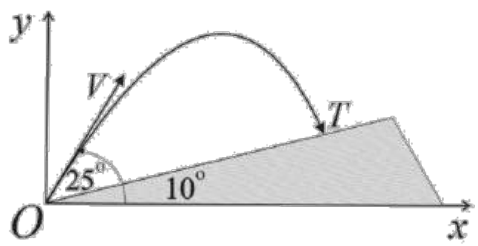
Note: Options A and B are out when you consider $\ddot{y}$.

The initial velocity vector is $\begin{pmatrix} 60\cos 30° \\ 60\sin 30° \end{pmatrix}$

$\dot{x} = 30\sqrt{3}, \ddot{x} = 0, \dot{y} = 30$ and $\ddot{y} = -10$

3 The diagram shows a road OT that makes an angle of 10^o with the horizontal.

A projectile is fired from O at an angle of 25^o to the horizontal, with initial velocity $V = 20\ m/s$. It hits a target at T. Assume the acceleration due to gravity is $10\ m/s^2$.



(i). Find the time taken to hit the target. 3

i) Let $\underset{\sim}{a} = -g\underset{\sim}{j}$ Then: $\ddot{x} = 0$ and $\ddot{y} = -g$; $V = 20$ m/s

[4] Finding velocity:

$\underset{\sim}{v} = \int \underset{\sim}{a}\, dt$

$= \int -10\underset{\sim}{j}\, dt$

$= -10t\underset{\sim}{j} + \underset{\sim}{C}$

Finding $\underset{\sim}{C}$:

$\underset{\sim}{v} = -10t\underset{\sim}{j} + \underset{\sim}{C}$ ∴ when $t = 0$, $\underset{\sim}{v} = \underset{\sim}{C}$

Also, initial velocity is $\underset{\sim}{v} = 20\cos\theta\underset{\sim}{i} + 20\sin\theta\underset{\sim}{j}$

∴ $\underset{\sim}{C} = v\cos\theta\underset{\sim}{i} + v\sin\theta\underset{\sim}{j}$

∴ $\underset{\sim}{v} = -10t\underset{\sim}{j} + 20\cos\theta\underset{\sim}{i} + 20\sin\theta\underset{\sim}{j}$

Now, $\underset{\sim}{s} = \int \underset{\sim}{v}\, dt$

$= \int [20\cos\theta\underset{\sim}{i} + (-10t + 20\sin\theta)\underset{\sim}{j}]\, dt$

$= 20t\cos\theta\underset{\sim}{i} + \left(-\frac{10t^2}{2} + 20t\sin\theta\right)\underset{\sim}{j} + \underset{\sim}{D}$

Initially when $t = 0$, $|\underset{\sim}{s}| = 0$ ∴ $\underset{\sim}{s} = 0\underset{\sim}{i} + 0\underset{\sim}{j}$

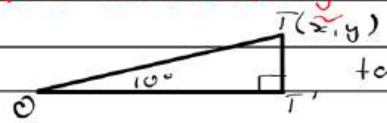
∴ $0 = 20\times0\times\cos\theta\underset{\sim}{i} + (-5\times0 + 20\times0\times\sin\theta)\underset{\sim}{j} + \underset{\sim}{D}$

∴ $\underset{\sim}{D} = \underset{\sim}{0}$

$\underset{\sim}{s} = 20t\cos\theta\underset{\sim}{i} + (-5t^2 + 20t\sin\theta)\underset{\sim}{j}$

∴ $\underset{\sim}{s} = 20t\cos25°\underset{\sim}{i} + (-5t^2 + 20t\sin25°)\underset{\sim}{j}$

Given the diagram:



$\tan 10° = \frac{y}{x}$

∴ $\tan 10° = \frac{-5t^2 + 20t\sin25°}{20t\cos25°}$

$= \frac{5t(4\sin25° - t)}{20t\cos25°}$

$= \frac{4\sin25° - t}{4\cos25°}$

∴ $t = 4\sin25° - 4\tan10°\cos25°$

$= 1.05124\ldots$ seconds

(ii). Find the distance OT. Give your answer correct to two decimal places. 1

ii) [2] $\cos 10° = \frac{OT'}{OT}$

$= \frac{x}{OT}$

∴ $OT = \frac{x}{\cos 10°}$

∴ $OT = \frac{20 \times 1.05124\ldots \times \cos25°}{\cos 10°}$

$= 19.34889\ldots$

$\doteqdot 19.35$ m

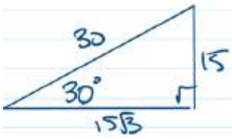
4 At a carnival, an air cannon fires a t-shirt at time $t = 0$ seconds from an origin O at ground level across a level field. The position vector $\underset{\sim}{r}(t) = 15\sqrt{3}t\underset{\sim}{i} + (15t - 4.9t^2)\underset{\sim}{j}$, where $\underset{\sim}{i}$ is a unit vector in the forward direction, $\underset{\sim}{j}$ is a unit vector vertically up and displacement components are measured in metres.

(i). Find the initial velocity of the t-shirt and the initial angle, in degrees, of its trajectory to the horizontal. 2

$\underset{\sim}{r}(t) = 15\sqrt{3}t\underset{\sim}{i} + (15t - 4.9t^2)\underset{\sim}{j}$

$\dot{\underset{\sim}{r}}(t) = 15\sqrt{3}\underset{\sim}{i} + (15 - 9.8t)\underset{\sim}{j}$

$\dot{\underset{\sim}{r}}(0) = 15\sqrt{3}\underset{\sim}{i} + 15\underset{\sim}{j}$.



initial velocity is $30\,ms^{-1}$, with an angle of $30°$.

(ii). Find the maximum height reached by the t-shirt, giving your answer in metres, correct to two decimal places. 2

max height when $\dot{y} = 0$

vertical velocity $\dot{y} = 15 - 9.8t$

$\Rightarrow t = 1.5306$

$y = 15t - 4.9t^2$

$= 15(1.5306) - 4.9(1.5306)^2$

$= 11.47959 = 11.48$ m

(iii). Find the time of flight of the t-shirt. Give your answer in seconds, correct to three decimal places. 1

time of flight when $y = 0$

$15t - 4.9t^2 = 0$

$t(15 - 4.9t) = 0$

$\therefore t = 3.061$ sec

(iv). Find the range of the t-shirt in metres, correct to one decimal place. 1

at $t = 3.061$,

$x = 15\sqrt{3}(3.061)$

$= 79.52711$

$= 79.5$ m

(v). A person in the crowd, more than 40 m from O, catches the t-shirt at a height of 2 m above the ground. How far horizontally from O is this person when the t-shirt is caught? Give your answer in metres, correct to one decimal place. 2

$y = 2$

$15t - 4.9t^2 = 2$

$4.9t^2 - 15t + 2 = 0$

$t = \frac{15 \pm \sqrt{125 - 4(4.9)2}}{2(4.9)}$

$= \frac{15 + \sqrt{85.8}}{9.8}$

$= 2.4758$ sec

$x = 15\sqrt{3} \times 2.4758$

$= 64.323$

$= 64.3$ m.

5 One Spring Tuesday morning in Term 4, at 8:30 am, Sam was on the DA Tuition hockey fields playing footy with the boys. He then kicked the football to David with all his might. If Sam kicked the football from O, on level ground with velocity 25 m/s at an inclination of θ, the equations of motion of the football are:

$$s(t) = 25t\cos\theta \underset{\sim}{i} + (25t\sin\theta - 4.9t^2)\underset{\sim}{j}$$

Unfortunately, Sam used too much power and the football smashed the window when it reached its maximum height.

(i). Show that the football smashes the window at a height of $\frac{3125}{98}\sin^2\theta$ m. 2

$s(t) = 25t\cos\theta \underset{\sim}{i} + (25t\sin\theta - 4.9t^2)\underset{\sim}{j}$

Football breaks window at max height.

Calculate max height when vertical velocity = 0.

$y = 25t\sin\theta - 4.9t^2$ (vertical displacement)

$\dot{y} = 25\sin\theta - 9.8t$

$0 = 25\sin\theta - 9.8t$

$9.8t = 25\sin\theta$

$t = \frac{25\sin\theta}{9.8}$

Sub t into vertical displacement.

$y = 25\left(\frac{25\sin\theta}{9.8}\right)\sin\theta - 4.9\left(\frac{25\sin\theta}{9.8}\right)^2$

$= \frac{625\sin^2\theta}{9.8} - \frac{3125\sin^2\theta}{98}$

$y = \frac{3125\sin^2\theta}{98}\sin^2\theta$ m

(ii). Show that the football hits the window at a horizontal distance of $\frac{3125}{98}\sin 2\theta$ m from O. 1

Sub t into horizontal displacement.

$x = 25\left(\frac{25\sin\theta}{9.8}\right)\cos\theta = \frac{25^2\sin\theta\cos\theta}{9.8}$

$= \frac{625}{9.8} \times \frac{1}{2}\sin 2\theta = \frac{3125}{98}\sin 2\theta$

6 A golfer hits a golf ball from a point O with speed $V\ m/s$ at an angle of 30° above the horizontal and moves freely under gravity. The ball reaches its greatest height at time T seconds after projection. The position vector, $\underset{\sim}{r}$, at any time, t seconds, is given by 3

$$\underset{\sim}{r}(t) = \begin{pmatrix} \frac{\sqrt{3}}{2}Vt \\ \frac{1}{2}Vt - 5t^2 \end{pmatrix}$$

Find, in terms of V, the speed of the ball at time $\frac{2}{3}T$ seconds after projection.

$$\therefore \dot{\underset{\sim}{r}}(t) = \begin{pmatrix} \frac{\sqrt{3}}{2}V \\ \frac{1}{2}V - 10t \end{pmatrix}$$

The ball reaches its greatest height when $t = T$ and $\dot{y} = 0$ i.e. $\frac{1}{2}V - 10T = 0$

$\therefore 10T = \frac{1}{2}V \qquad \therefore T = \frac{1}{20}V$

So at $t = \frac{2}{3}T = \frac{1}{30}V$: $\qquad \dot{\underset{\sim}{r}}\left(\frac{1}{30}V\right) = \begin{pmatrix} \frac{\sqrt{3}}{2}V \\ \frac{1}{2}V - 10 \times \frac{1}{30}V \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2}V \\ \frac{1}{6}V \end{pmatrix}$

The speed is $\left|\dot{\underset{\sim}{r}}\left(\frac{1}{30}V\right)\right|$ where $\left|\dot{\underset{\sim}{r}}\left(\frac{1}{30}V\right)\right| = \sqrt{\left(\frac{\sqrt{3}}{2}V\right)^2 + \left(\frac{1}{6}V\right)^2} = \frac{\sqrt{7}}{9}V$ m/s

7 A particle, experiencing vertical acceleration due to gravity g and no air resistance, is projected over horizontal ground at speed $v\ m/s$ at an acute angle θ to the horizontal. You may assume the six equations of motion listed below.

$$\ddot{y} = -g\underset{\sim}{j} \qquad \ddot{x} = 0$$

$$\dot{y} = (-gt + vsin\theta)\ \underset{\sim}{j} \qquad \dot{x} = vcos\theta\underset{\sim}{i}$$

$$y = (-\frac{g}{2}t^2 + vtsin\theta)\underset{\sim}{j} \qquad x = vtcos\theta\underset{\sim}{i}$$

(i). Shat that the

α) time to reach greatest height is $\frac{vsin\theta}{g}$. 1

Greatest height occurs when

$0 = \dot{y}$

$0 = -gt + v\sin\theta$

$gt = v\sin\theta$

$t = \frac{v\sin\theta}{g}$

β) greatest height is $H = \frac{v^2\,sin^2\,\theta}{2g}$. 1

When $t = \frac{v\sin\theta}{g}$, $y = H$

$H = -\frac{g}{2}\left(\frac{v\sin\theta}{g}\right)^2 + v\left(\frac{v\sin\theta}{g}\right)\sin\theta$

$= -\frac{v^2\sin^2\theta}{2g} + \frac{v^2\sin^2\theta}{g} = \frac{v^2\sin^2\theta}{2g}$

γ) time to reach the landing point is $\frac{2vsin\theta}{g}$. 1

When $y = 0$,

$0 = -\frac{gt^2}{2} + vt\sin\theta$

$0 = t\left(-\frac{gt}{2} + v\sin\theta\right)$

$t = 0$ or $\frac{2v\sin\theta}{g}$

time to landing point is $\frac{2v\sin\theta}{g}$

δ) range is $R = \frac{v^2 sin2\theta}{g}$. 1

when $t = \frac{2v\sin\theta}{g}$, Range $= x$

$\therefore$ Range $= v\left(\frac{2v\sin\theta}{g}\right)\cos\theta$

$= \frac{v\sin 2\theta}{g}$

(ii). If R is three times H, find the exact angle of projection. 2

$R = 3H$

$\frac{v^2\sin 2\theta}{g} = \frac{3v^2\sin^2\theta}{2g}$

$\sin 2\theta = \frac{3}{2}\sin^2\theta$

$2\sin\theta\cos\theta = \frac{3}{2}\sin^2\theta$

$4\sin\theta\cos\theta - 3\sin^2\theta = 0$

$\sin\theta(4\cos\theta - 3\sin\theta) = 0$

$\sin\theta = 0$ or $4\cos\theta = 3\sin\theta$

$\theta = 0$ $\quad$ $4/3 = \tan\theta$

$\theta = \tan^{-1}(4/3)$

$\therefore$ Exact angle of projection is $\tan^{-1}(4/3)$